

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Department of Mechanical Engineering
IM0240 – Machine Design I

**Design proposal for a
CNC router machine**
Tripteron 3-PRRR architecture
Delivery 1 – Corrected English edition

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1. Introduction

Within the Machine Design course, the assignment is to conceive, design and build a CNC router that satisfies a specific set of technical requirements. The goal is a machine that, using the concepts studied in the course, can be adapted to the manufacturing processes where light routing and engraving are needed.

This document is the first delivery of the project. It defines the problem, surveys the state of the art, establishes the requirements and the product design specification, compares the mechanism architectures that could satisfy them and justifies the architecture that was selected: a **Tripteron**, a three-degree-of-freedom translational parallel machine with three PRRR legs actuated by power screws.

This is a corrected and restructured English edition of the original proposal. The technical content of the original has been preserved; the errors identified in it are listed and corrected in Section 12.

2. Brief

2.1. What is a router machine?

A CNC router is an automated machine that uses computer numerical control to cut, carve or engrave materials such as wood, plastic, metal and composites. It works with a rotating cutting tool — an end mill — whose path is controlled by a program. Because the process is both precise and repeatable, routers are a basic tool in manufacturing and design workshops.

2.2. How does it work?

The tool moves in a three-dimensional coordinate system driven by three motors. The operator's controller commands the motors so that the tool sweeps the programmed path inside the working volume. Each motor drives one power screw, and each power screw moves one axis of the machine.

2.3. Which materials are viable for its construction?

Different parts call for different materials:

Part	Material	Reason
External enclosure	PVC or wood	Carries no significant load; low cost, easy to shape
Structural frame	Steel or aluminium	Carries the machining loads and the weight of the drives
Mechanism (links, screws)	Steel	Stiffness and wear resistance govern the accuracy

3. Objectives

3.1. General objective

Design and build a numerically controlled router able to cut at least 1 mm of depth in a range of materials, making the best use of the available materials, manufacturing processes and aesthetics.

3.2. Specific objectives

- Define the flows through the machine — the inputs it needs and the outputs it must deliver — so that the concept is built on a solid functional basis.
- Develop a support structure that guarantees stable operation, minimising vibration and avoiding buckling under the weight of the equipment.
- Design a machine that meets the client's expectations in appearance, design and aesthetics, so that the result is attractive from the user's point of view.
- Plan the specific requirements needed to satisfy the client, ensuring the technical and functional feasibility of the project.
- Select a kinematic chain and a mechanism that suit both the client's needs and an efficient development process, guaranteeing adequate performance.

4. State of the art

4.1. CNC machines

A CNC (computer numerical control) machine removes the human factor from the trajectory of the tool: the cut is performed automatically by a computer that feeds numerical coordinates to the machine, which is what makes millimetre and sub-millimetre precision repeatable.

4.2. Motion control

The motion of a CNC machine is controlled by software that interprets G-code. The general process is:

1. **Model design.** The part is modelled in 3D with a CAD package.
2. **G-code generation.** A CAM package converts the model into the G-code that specifies tool movements, cutting speeds and tooling.
3. **Transfer.** The program is transferred to the machine over a USB drive or a network connection.
4. **Set-up.** The tool is mounted, the stock is clamped and the cutting parameters — feed rate and spindle speed — are configured.
5. **Execution.** The controller interprets the G-code and moves the axes accordingly, producing the part.

4.3. The CNC program

A CNC program is a set of coded instructions written mainly in G-code, complemented by M, S and T codes depending on the functions the machine offers.

4.4. Types of CNC machines

Criterion	Classes
By function	Milling machines, lathes, drilling machines, plasma cutters, grinders.
By motion	<i>Point to point</i> : the stock and the tool are positioned and held while the cutter completes the operation, then retracted. <i>Contouring</i> : the tool follows the contour of the part along a continuous path.
By number of axes	2 axes; 2.5 axes (the three axes move, but x and y position first and only then does z work); 3 axes; 4 axes (3 axes plus one rotation, A or B); 5 axes (3 axes plus two rotations, A and B).

The machine of this project is a **3-axis contouring router**.

4.5. Applications

Manufacture of metal and plastic parts; cutting and engraving of wood, metal, plastic and stone for signage, architectural models and artistic engraving; prototyping and mass production of identical parts; medical devices, prostheses and surgical instrument components; and, on machines equipped for it, additive manufacturing.

4.6. Key components

Power screw

A mechanical device that converts rotary into linear motion. A threaded screw engages a nut with an internal thread; turning the screw translates the nut along its axis. It is the element that turns the rotation of the motor into the feed of the axis, and it is sized in Delivery 3.

NEMA 34 stepper motor

A NEMA standard frame size, roughly 86 mm across the body, known for high torque. It is used in industrial CNC machines, large-format 3D printers and automation equipment. The closed-loop 86HB250 variant selected for this project delivers 4.5 N m of holding torque.

End mill

The rotating cutting tool. A sharpened metal tip turns at high speed and removes material on contact. End mills are interchangeable and are chosen according to the material and the finish required.

5. User definition

The user of the router is someone experienced with cutting equipment who is looking for a versatile, easy to operate and affordable machine. The intended duty is light cutting without extreme demands, in a workshop with a mains supply. The machine is to be operated by people of at least 18 years of age.

6. Conceptual design

6.1. Requirements imposed by the course

- A cutting machine (router) with **3 degrees of freedom**.
- The machine **may not** be a cartesian system built from prismatic pairs in series.
- The degrees of freedom must be **coupled**, i.e. the architecture must be a parallel one.
- The prismatic pairs must be **actuated by power screws**.
- The machine may be spatial (linear delta or Tripteron) or planar.
- It must be able to cut at least 1 mm deep in aluminium, copper, brass or nylon.

6.2. Technical design requirements

Requirement	Definition
Feed rate	Constant routing speed of approximately 10 mm/s.
Accuracy	Path error not greater than 1 mm.
Cutting capability	Able to route wood, aluminium, copper, brass and nylon to a depth of at least 1 mm.
Safety	The machine must not put the operator at risk during operation.
Control	The operator must have buttons or instruments to start and stop the process.
Energy and information	The machine must carry a control system that transmits accurate process information while a stable power supply is guaranteed.

6.3. Black box

The black box states *what* the machine does without saying *how*: the three flows — material, energy and information — that cross its boundary.

6.4. Functional structure

The functional structure — the “transparent box” — decomposes the overall function into the elementary functions that the machine must perform, which is what allows the concept to be generated function by function instead of by copying an existing machine.

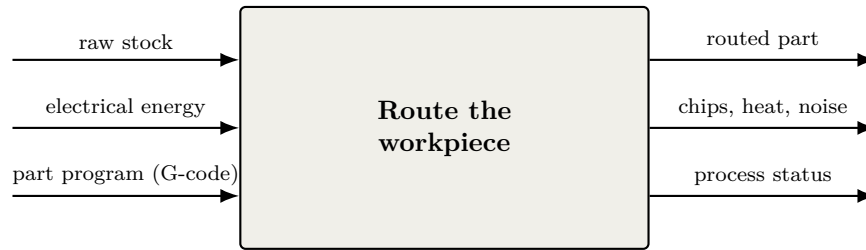


Figure 1: Black box of the router: material, energy and information flows.

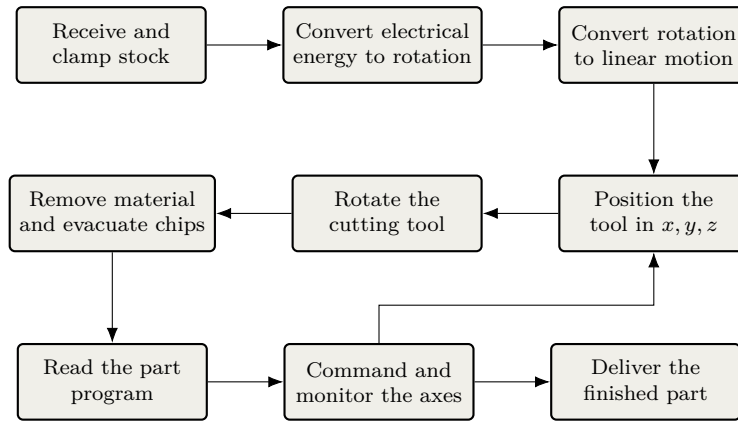


Figure 2: Functional structure of the machine.

6.5. Attributes

The attributes are the measurable quantities through which each flow is evaluated in the product design specification.

Material	On/off	Manual work	Energy
Strength	Production accuracy	Accuracy	Transfer
Specific weight	Power required	Instructions	Voltage
Size		Ease of production	Current
Thickness			Power

6.6. Product design specification (PDS)

#	Requirement	Metric	Value	Unit	Weight	D/d
1	Degrees of freedom	Coupled DOF	3	DOF	4	D
2	Service life	Time	3	h	4	d
3	Ease of assembly	Time	< 3	h	4	D
4	Ease of maintenance	Time	2	h	4	d
5	Cost	Money	500 000	COP	5	D
6	Construction time	Time	3	months	4	d
7	Width	Length	0.55	m	5	d
8	Depth	Length	0.55	m	5	d
9	Height	Length	0.55	m	5	d
10	Feed rate	Speed	10	mm/s	3	D
11	Path accuracy	Length	≤ 1	mm	5	D
12	Depth of cut	Length	≥ 1	mm	5	D
13	Emergency stop response	Time	0.5	s	6	D
14	Positioning resolution	Length	0.1	mm	7	D

D = demand (must be met), d = desire (desirable). The units of rows 10–14 were corrected with respect to the original proposal; see Section 12.

6.7. Morphological matrix

Each elementary function of the functional structure was given a set of possible solutions, and the concepts were generated by combining one solution per function.

Function	Solution A	Solution B	Solution C
Convert energy to rotation	Stepper motor	Servo motor	DC motor with encoder
Convert rotation to translation	Power screw	Belt and pulley	Rack and pinion
Guide the carriage	Linear rail and bearing block	Round shaft and bushing	Dovetail slide
Mechanism architecture	Tripteron (3-PRRR)	Linear delta	Serial cartesian (not allowed)
Structure	Welded steel tube frame	Bolted aluminium extrusion	Laser-cut plate
Control	Microcontroller with drivers	Industrial CNC controller	PC with parallel port

The evaluation of the resulting combinations against the PDS selected: stepper motors, power screws, linear rails, a welded steel frame, a microcontroller-based controller, and the **Tripteron** architecture.

7. Mechanism architecture

7.1. Topological synthesis

Base

The fixed frame. It does not move while the machine is operating and it carries the three guides, the

three screws and the work table.

Arms

Three identical legs. Each leg is driven along one axis and transmits that motion to the platform through two links and three parallel revolute joints.

Cutting tool

Mounted on the platform where the three legs meet. It performs the milling, drilling or cutting work.

7.2. Mobility

The machine has 10 moving bodies — three carriages, six links and the platform — and 12 joints, each with one degree of freedom (3 prismatic and 9 revolute). The Kutzbach–Grübler criterion for spatial mechanisms gives

$$M = 6(n_b) - \sum_i (6 - f_i) = 6 \times 10 - 12 \times 5 = 0, \quad (1)$$

where $n_b = 10$ is the number of moving bodies and $f_i = 1$ for every joint.

The formula predicts a structure, yet the machine moves with three degrees of freedom. The discrepancy is not an error: the Tripteron belongs to the family of **over-constrained** mechanisms, in which the special geometry — the three revolute axes of each leg parallel to one another and perpendicular to its prismatic axis — makes three of the 60 constraint equations redundant. The true mobility is therefore

$$M = 0 + 3 = 3 \quad \text{translations, no rotation.} \quad (2)$$

Exactly the same number of redundancies appears again in Delivery 3, where the static equilibrium system turns out to have 63 unknowns and rank 60. The two analyses are consistent, and the consequence for manufacturing is important: an over-constrained machine is sensitive to assembly errors, so the parallelism and squareness tolerances of the three guides must be tight.

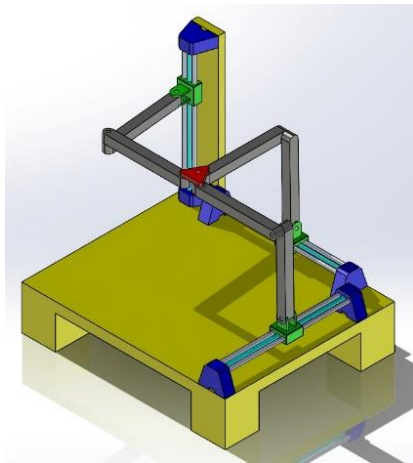
7.3. Why a Tripteron

Architecture	Advantages	Disadvantages
Serial cartesian	Simple, decoupled, cubic workspace	Not allowed by the course; each axis carries the next
Linear delta	Light moving mass, fast	Coupled kinematics, curved workspace boundary, singularities inside the useful volume
Tripteron	Decoupled like a cartesian machine but built as a parallel mechanism; cubic workspace; actuators fixed to the frame; simple controller	Over-constrained, so it demands accurate assembly; more links than a delta

The Tripteron was selected because it satisfies the course requirement of coupled, parallel architecture while keeping the two practical advantages of a cartesian machine: a cubic workspace and a controller that needs no coordinate transformation. Delivery 3 confirms both properties numerically.

8. Concept proposals

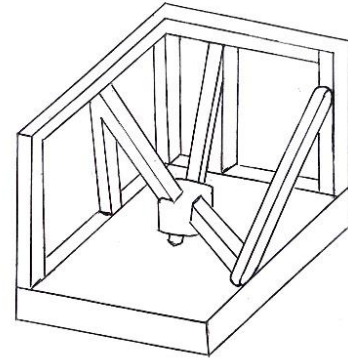
Each member of the team developed an individual concept; the concepts were then compared against the PDS and merged into a single final concept.



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Figure 3: Individual concepts.

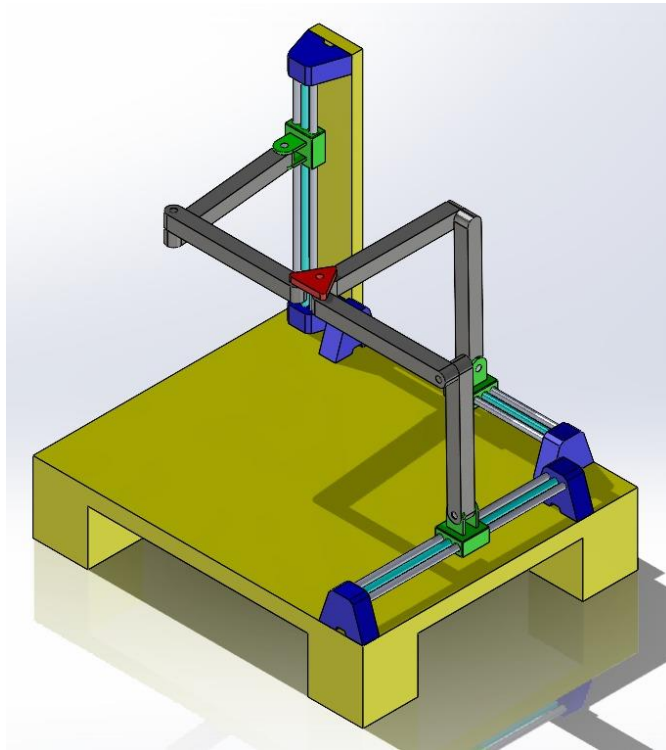


Figure 4: Final concept: welded frame, three orthogonal screw-driven axes, Tripterion legs and a spindle mounted on the moving platform.

The final concept keeps the vertical axis on a column at the rear of the frame and the two horizontal axes along the base, so that the three screws are all mounted on the fixed structure and no motor travels with the tool. The work table is fixed, which removes the need to accelerate the workpiece.

9. Work methodology

The machine is developed collaboratively: the drawing package is produced jointly, and the construction of the enclosure, structure and mechanism is split among specialised sub-groups whose work is integrated at the end. The team meets weekly to agree on the next tasks and to report on the ones already completed.

Milestone	Delivery	Content
1	Delivery 1	Concept, requirements, PDS, architecture selection
2	Delivery 2	Complete drawing package and bill of materials
3	Delivery 3	Calculation report: kinematics, statics, transmission, FEA
4	Final	Manufactured and assembled machine

10. Planned calculations

- A **kinematic analysis** to obtain the position, velocity and acceleration of every body of the machine, and to establish the workspace.
- A **static analysis** to identify the forces on the actuators, the joints and the links.
- A **solid mechanics** verification to confirm that the selected materials withstand those forces — in particular the power screws, which are slender and loaded in compression.
- A **finite element analysis** of the frame, whose redundant three-dimensional structure cannot be resolved by hand.

All four were carried out and are reported in Delivery 3.

11. Conclusions

1. A Tripteron was selected over a linear delta because it is the only parallel architecture that keeps the decoupled behaviour and the cubic workspace of a cartesian machine, which simplifies both the control and the programming of the part.
2. The design must be optimised for material usage and cost: the PDS sets a budget of COP 500 000 and an overall size of 0.55 m per side, and both are binding constraints on the frame.
3. The mobility analysis shows that the machine is over-constrained. This is acceptable, but it converts the parallelism of the three guides into a critical manufacturing tolerance, and it must be watched during assembly.
4. The kinematic analysis is required before the structure can be dimensioned, since the stroke of each screw and the length of the links determine the size of the frame.

12. Corrections with respect to the original proposal

#	Original	Problem	Correction
1	Mobility computed as $M = 3n - 2J_1 - J_2 = 3$	That is Grübler’s formula for <i>planar</i> mechanisms; the Tripteron is spatial.	Kutzbach for spatial mechanisms, $M = 6n_b - \sum(6 - f_i) = 0$, plus 3 redundant constraints $\Rightarrow M = 3$
2	“3 DOF: rotation about the base (z) and horizontal motion”	The Tripteron platform does not rotate; its three degrees of freedom are pure translations.	3 translations x, y, z ; zero rotations
3	Feed rate listed as “300 millimetres” with the metric “distance”	A feed rate is a speed, not a distance; the value contradicted the 10 mm/s stated in the text.	10 mm/s
4	Accuracy listed as “85 percent”	Accuracy of a machine tool is a length, not a percentage.	Path error ≤ 1 mm
5	Cutting capability listed as “10 cm/s ² ”, metric “area/time ² ”	Neither the unit nor the metric corresponds to a depth of cut.	Depth of cut ≥ 1 mm
6	Service life of 3 hours	Three hours is a test duration, not a service life.	Retained as stated, but reinterpreted as continuous operating time between maintenance stops
7	Section numbering out of step (section 3 containing subsections 4.1, 4.2, . . .)	Made cross-referencing impossible.	Numbering rebuilt
8	Repeated figures and an unattributed image taken from the internet	Duplicated content and unreferenced third-party material.	Duplicates removed; only the team’s own material is reproduced

13. References

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